

FOSD — Unit 3

Synchronization & Concurrency

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- The selected question numbers are listed below; neighboring questions may remain visible because the original page is preserved.
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Included source pages

CS2018 — source page 17 — selected Q40
CS2020 — source page 11 — selected Q34
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CS2022 — source page 15 — selected Q19
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- Q.40 Consider the following solution to the producer-consumer synchronization problem. The shared buffer size is N . Three semaphores *empty*, *full* and *mutex* are defined with respective initial values of 0, N and 1. Semaphore *empty* denotes the number of available slots in the buffer, for the consumer to read from. Semaphore *full* denotes the number of available slots in the buffer, for the producer to write to. The placeholder variables, denoted by P , Q , R , and S , in the code below can be assigned either *empty* or *full*. The valid semaphore operations are: `wait()` and `signal()`.

Producer:	Consumer:
<pre>do{ wait(P); wait(mutex); //Add item to buffer signal(mutex); signal(Q); }while(1);</pre>	<pre>do{ wait(R); wait(mutex); //Consume item from buffer signal(mutex); signal(S); }while(1);</pre>

Which one of the following assignments to P , Q , R and S will yield the correct solution?

- (A) P : *full*, Q : *full*, R : *empty*, S : *empty*
 (B) P : *empty*, Q : *empty*, R : *full*, S : *full*
 (C) P : *full*, Q : *empty*, R : *empty*, S : *full*
 (D) P : *empty*, Q : *full*, R : *full*, S : *empty*
- Q.41 Consider the relations $r(A, B)$ and $s(B, C)$, where $s.B$ is a primary key and $r.B$ is a foreign key referencing $s.B$. Consider the query

$$Q: r \bowtie (\sigma_{B < 5}(s))$$

Let *LOJ* denote the natural left outer-join operation. Assume that r and s contain no null values.

Which one of the following queries is NOT equivalent to Q ?

- (A) $\sigma_{B < 5}(r \bowtie s)$ (B) $\sigma_{B < 5}(r \text{ LOJ } s)$
 (C) $r \text{ LOJ } (\sigma_{B < 5}(s))$ (D) $\sigma_{B < 5}(r) \text{ LOJ } s$

Q.No. 32 Consider the following languages.

$$L_1 = \{wxyx \mid w, x, y \in (0 + 1)^+\}$$

$$L_2 = \{xy \mid x, y \in (a + b)^*, |x| = |y|, x \neq y\}$$

Which one of the following is TRUE?

- (A) L_1 is regular and L_2 is context-free.
- (B) L_1 is context-free but not regular and L_2 is context-free.
- (C) Neither L_1 nor L_2 is context-free.
- (D) L_1 is context-free but L_2 is not context-free.

Q.No. 33 Consider the productions $A \rightarrow PQ$ and $A \rightarrow XY$. Each of the five non-terminals A, P, Q, X, and Y has two attributes: s is a synthesized attribute, and i is an inherited attribute. Consider the following rules.

Rule 1: $P.i = A.i + 2$, $Q.i = P.i + A.i$, and $A.s = P.s + Q.s$

Rule 2: $X.i = A.i + Y.s$ and $Y.i = X.s + A.i$

Which one of the following is TRUE?

- (A) Both Rule 1 and Rule 2 are L-attributed.
- (B) Only Rule 1 is L-attributed.
- (C) Only Rule 2 is L-attributed.
- (D) Neither Rule 1 nor Rule 2 is L-attributed.

Q.No. 34 Each of a set of n processes executes the following code using two semaphores a and b initialized to 1 and 0, respectively. Assume that `count` is a shared variable initialized to 0 and not used in CODE SECTION P.

CODE SECTION P

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wait(a); count=count+1;
if (count==n) signal(b);
signal(a); wait(b); signal(b);
```

CODE SECTION Q

What does the code achieve?

- (A) It ensures that no process executes CODE SECTION Q before every process has finished CODE SECTION P.
- (B) It ensures that at most two processes are in CODE SECTION Q at any time.
- (C) It ensures that all processes execute CODE SECTION P mutually exclusively.
- (D) It ensures that at most $n-1$ processes are in CODE SECTION P at any time.

Q.No. 35



Computer Science and Information Technology (CS, Set-1)

Q.46	Consider the following pseudocode, where S is a semaphore initialized to 5 in line#2 and counter is a shared variable initialized to 0 in line#1. Assume that the increment operation in line#7 is <i>not</i> atomic. <pre> 1. int counter = 0; 2. Semaphore S = init(5); 3. void parop(void) 4. { 5. wait(S); 6. wait(S); 7. counter++; 8. signal(S); 9. signal(S); 10.}</pre> If five threads execute the function parop concurrently, which of the following program behavior(s) is/are possible?
(A)	The value of counter is 5 after all the threads successfully complete the execution of parop .
(B)	The value of counter is 1 after all the threads successfully complete the execution of parop .
(C)	The value of counter is 0 after all the threads successfully complete the execution of parop .
(D)	There is a deadlock involving all the threads.

GATE 2022 Computer Science and Information Technology (CS)

Q.19	Consider the following threads, T_1 , T_2 , and T_3 executing on a single processor, synchronized using three binary semaphore variables, S_1 , S_2 , and S_3 , operated upon using standard <code>wait()</code> and <code>signal()</code> . The threads can be context switched in any order and at any time.		
	T_1 <pre>while(true){ wait(S_3); print("C"); signal(S_2); }</pre>	T_2 <pre>while(true){ wait(S_1); print("B"); signal(S_3); }</pre>	T_3 <pre>while(true){ wait(S_2); print("A"); signal(S_1); }</pre>
	Which initialization of the semaphores would print the sequence BCABCABCA....?		
(A)	$S_1 = 1; S_2 = 1; S_3 = 1$		
(B)	$S_1 = 1; S_2 = 1; S_3 = 0$		
(C)	$S_1 = 1; S_2 = 0; S_3 = 0$		
(D)	$S_1 = 0; S_2 = 1; S_3 = 1$		

Q.38	Consider the two functions <code>incr</code> and <code>decr</code> shown below. <pre> incr(){ wait(s); X = X+1; signal(s); } incr(){ wait(s); X = X+1; signal(s); } decr(){ wait(s); X = X-1; signal(s); } decr(){ wait(s); X = X-1; signal(s); }</pre> There are 5 threads each invoking <code>incr</code> once, and 3 threads each invoking <code>decr</code> once, on the same shared variable <code>X</code>. The initial value of <code>X</code> is 10. Suppose there are two implementations of the semaphore <code>s</code>, as follows: I-1: <code>s</code> is a binary semaphore initialized to 1. I-2: <code>s</code> is a counting semaphore initialized to 2. Let <code>V1</code>, <code>V2</code> be the values of <code>X</code> at the end of execution of all the threads with implementations I-1, I-2, respectively. Which one of the following choices corresponds to the minimum possible values of <code>V1</code>, <code>V2</code>, respectively?
(A)	15, 7
(B)	7, 7
(C)	12, 7
(D)	12, 8

- Q.46 Consider a multi-threaded program with two threads T1 and T2. The threads share two semaphores: s1 (initialized to 1) and s2 (initialized to 0). The threads also share a global variable x (initialized to 0). The threads execute the code shown below.

<pre>// code of T1 wait(s1); x = x+1; print(x); wait(s2); signal(s1);</pre>	<pre>// code of T2 wait(s1); x = x+1; print(x); signal(s2); signal(s1);</pre>
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Which of the following outcomes is/are possible when threads T1 and T2 execute concurrently?

- (A) T1 runs first and prints 1, T2 runs next and prints 2
- (B) T2 runs first and prints 1, T1 runs next and prints 2
- (C) T1 runs first and prints 1, T2 does not print anything (deadlock)
- (D) T2 runs first and prints 1, T1 does not print anything (deadlock)

- Q.47 Let A be an $n \times n$ matrix over the set of all real numbers $\mathbb{R}$. Let B be a matrix obtained from A by swapping two rows. Which of the following statements is/are TRUE?

- (A) The determinant of B is the negative of the determinant of A
- (B) If A is invertible, then B is also invertible
- (C) If A is symmetric, then B is also symmetric
- (D) If the trace of A is zero, then the trace of B is also zero